

4

Strong coupling: self-trapping

This chapter presents the theory of an electron's self-trapping. The discussion begins by reviewing the adiabatic formalism. This formalism provides a method of describing an electron amidst atoms with which it interacts. In Section 4.2 the formalism is applied to the nonlinear problem of an electron's self-localization. A scaling approach is then employed in Section 4.3 to analyze an electron's self-trapping in the adiabatic limit in which atoms' masses are assumed to be infinite. Two distinct types of self-trapped state, corresponding to large- and small-polaron formation, are possible. The situations in which large and small polarons can form are described. The discussion proceeds beyond the adiabatic limit in Section 4.4 to describe the effects of atomic vibrations on self-trapped carriers and on the conditions of their formation. The chapter concludes by describing how disorder can induce electrons to collapse into small polarons.

4.1 Adiabatic formalism

An electron is very much lighter than an atom. This basic feature generates a qualitative asymmetry in the responses of an electronic carrier and a solid's atoms to their mutual interactions. The adiabatic approach exploits this asymmetry to describe the motion of an electronic carrier in a solid (Born and Oppenheimer, 1927).

The Hamiltonian for the added electron and the solid is written as

$$H \equiv H_e(\mathbf{r}, \mathbf{\Delta}) + H_{\text{vib}}(\mathbf{\Delta}). \quad (4.1)$$

The electronic portion of this Hamiltonian, $H_e(\mathbf{r}, \mathbf{\Delta})$, describes the carrier's kinetic energy and the potential arising from the carrier's interaction with the solid's atoms. The carrier's position is designated by $\mathbf{r}$, and $\mathbf{\Delta}(\mathbf{u})$ represents the displacement of the atom whose equilibrium position in a carrier-free solid is $\mathbf{u}$. Here, for notational simplicity, the set of atomic displacements is collectively represented by the single

atomic-deformation function $\Delta \equiv \Delta(\mathbf{u})$. The second contribution to the total Hamiltonian, $H_{\text{vib}}(\Delta)$, describes the kinetic energy of the atoms and their interactions with one another. In polaron studies this Hamiltonian usually describes harmonic vibrations of a solid's atoms.

The adiabatic approach is based on the eigenstates of an electron moving amongst atoms whose positions are fixed:

$$H_e(\mathbf{r}, \Delta)\phi_n(\mathbf{r}, \Delta) = W_n(\Delta)\phi_n(\mathbf{r}, \Delta). \quad (4.2)$$

Here $\phi_n(\mathbf{r}, \Delta)$ and $W_n(\Delta)$ represent the n -th electronic eigenstate and corresponding energy for an electron in a medium whose atomic-displacement vectors are collectively designated as Δ . However, the excess electron also affects atomic motion. In particular, atomic motion is described by the projection of the total Hamiltonian on the electronic state $\phi_n(\mathbf{r}, \Delta)$:

$$[\int d\mathbf{r} \phi_n^*(\mathbf{r}, \Delta) H \phi_n(\mathbf{r}, \Delta)] \chi_m^n(\Delta) = E_{n,m} \chi_m^n(\Delta), \quad (4.3)$$

where $\chi_m^n(\Delta)$ and $E_{n,m}$ depict the m -th vibration wavefunction and the corresponding total energy when the electron is in its n -th electronic state.

The left-hand side of Eq. (4.3) can be evaluated further with

$$H_{\text{vib}}(\Delta) = \frac{\hat{\mathbf{P}}^2}{2M} + V_{\text{vib}}(\Delta), \quad (4.4)$$

where $\hat{\mathbf{P}} \equiv (\hbar/i)\partial/\partial\Delta$ represents the atomic momentum operator and M is the appropriate reduced atomic mass. After some manipulations, the Hamiltonian governing atoms' motion can be expressed as (Emin and Holstein, 1969):

$$H_{\text{ad}} = \frac{\left[\hat{\mathbf{P}}^2 + q \left(\mathbf{A} \cdot \hat{\mathbf{P}} + \hat{\mathbf{P}} \cdot \mathbf{A} \right) - \hbar^2 \text{Re} \int d\mathbf{r} \phi_n^*(\mathbf{r}, \Delta) \frac{\partial^2 \phi_n(\mathbf{r}, \Delta)}{\partial \Delta^2} \right]}{2M} + V_{\text{vib}}(\Delta) + W_n(\Delta), \quad (4.5)$$

where a portion of the Hamiltonian's inertial contribution has been written in terms of a fictitious vector potential for an electronic carrier of charge q :

$$q\mathbf{A}(\Delta) \equiv \frac{\hbar}{i} \int d\mathbf{r} \phi_n^*(\mathbf{r}, \Delta) \frac{\partial \phi_n(\mathbf{r}, \Delta)}{\partial \Delta}. \quad (4.6)$$

The final term in the square brackets is usually dismissed as being relatively small.

In the absence of a magnetic field a non-degenerate electronic state can be taken to be real. The fictitious vector potential then vanishes since

$$\int d\mathbf{r} \phi_n(\mathbf{r}, \Delta) \frac{\partial \phi_n(\mathbf{r}, \Delta)}{\partial \Delta} = \frac{1}{2} \frac{\partial}{\partial \Delta} \int d\mathbf{r} \phi_n^2(\mathbf{r}, \Delta) = 0. \quad (4.7)$$

With a magnetic field $\phi_n(\mathbf{r}, \Delta)$ becomes complex and $A(\Delta)$ no longer vanishes. The effective magnetic field obtained from $A(\Delta)$ is proportional to the magnetic field applied to the electron. The adiabatic Hamiltonian of Eq. (4.5) then describes how the circulation of the electron in a magnetic field affects atoms' vibrations. Such considerations are central to the adiabatic theory of the polaron Hall effect (Emin and Holstein, 1969).

An electron can move adiabatically in response to atomic motion in accord with $\phi_n(\mathbf{r}, \Delta)$. However, atomic motion can also induce transitions between adiabatic states: $\phi_n(\mathbf{r}, \Delta) \rightarrow \phi_{n'}(\mathbf{r}, \Delta)$. The matrix element that governs such non-adiabatic transitions is

$$\frac{\hat{\mathbf{P}}}{M} \cdot \frac{\hbar}{i} \int d\mathbf{r} \phi_{n'}^*(\mathbf{r}, \Delta) \frac{\partial \phi_n(\mathbf{r}, \Delta)}{\partial \Delta} - \frac{\hbar^2}{2M} \int d\mathbf{r} \phi_{n'}^*(\mathbf{r}, \Delta) \frac{\partial^2 \phi_n(\mathbf{r}, \Delta)}{\partial \Delta^2}. \quad (4.8)$$

The first of these two terms typically dominates. Its magnitude is the product of an atomic velocity and $\hbar$ divided by the distance atoms must move to appreciably alter the electronic state ($\sim t/F$ with a short-range electron–phonon interaction). Non-adiabatic transitions only become important when the product of the relevant atomic velocity and the electronic matrix element exceeds $W_{n'}(\Delta) - W_n(\Delta)$. In most semiconducting materials non-adiabatic transitions can be ignored since atoms' vibration energies are usually much smaller than the relevant electronic energies. Thus the adiabatic approach is frequently adopted.

4.2 Self-trapping

The adiabatic approach is widely utilized to address self-trapping. Here the adiabatic method is used to study the self-trapping of an electron added to a deformable continuum. A linear electron–phonon interaction is introduced by taking the electron's potential energy to be proportional to the medium's deformations. These deformations, like a solid's dilation, are taken as scalar parameters attributed to positions within the continuum that are denoted by the vector $\mathbf{u}$. The continuum's strain energy is a quadratic function of these deformations.

The adiabatic potential for the electron and its deformable continuum is the sum of the electron's energy and the strain energy of the continuum. Finding the

minimum of the adiabatic energy involves calculating the first partial derivative of the electron's energy with respect to the deformation parameters, the $\Delta(\mathbf{u})$. This differentiation is equivalent to calculating the expectation value of the derivative of the Hamiltonian with respect to that deformation:

$$\begin{aligned}\frac{\partial W_n}{\partial \Delta} &= \frac{\partial}{\partial \Delta} \langle \phi_n(\mathbf{r}, \Delta) | H_e | \phi_n(\mathbf{r}, \Delta) \rangle \\ &= W_n \frac{\partial}{\partial \Delta} \langle \phi_n(\mathbf{r}, \Delta) | \phi_n(\mathbf{r}, \Delta) \rangle + \langle \phi_n(\mathbf{r}, \Delta) | \left| \frac{\partial H_e}{\partial \Delta} \right| \phi_n(\mathbf{r}, \Delta) \rangle \\ &= \langle \phi_n(\mathbf{r}, \Delta) | \left| \frac{\partial H_e}{\partial \Delta} \right| \phi_n(\mathbf{r}, \Delta) \rangle,\end{aligned}\quad (4.9)$$

where the normalization condition $\langle \phi_n(\mathbf{r}, \Delta) | \phi_n(\mathbf{r}, \Delta) \rangle = 1$ has been utilized.

The Hamiltonian for the electron in the deformable continuum is

$$H_e = \frac{\hbar^2}{2m} \nabla_r^2 + \int d\mathbf{u} Z(\mathbf{r} - \mathbf{u}) \Delta(\mathbf{u}), \quad (4.10)$$

where m is the electron's mass and $Z(\mathbf{r} - \mathbf{u})$ describes the magnitude and range of the electron–phonon interaction. The partial derivative of the electronic energy with respect to $\Delta(\mathbf{u})$ is therefore

$$\frac{\partial W_n}{\partial \Delta(\mathbf{u})} = \int d\mathbf{r} |\phi_n(\mathbf{r}, \Delta)|^2 Z(\mathbf{r} - \mathbf{u}). \quad (4.11)$$

The strain energy of the carrier-free harmonic deformable continuum is

$$V_{\text{strain}} = \frac{S}{2} \int d\mathbf{u} \Delta^2(\mathbf{u}), \quad (4.12)$$

where S is the continuum's stiffness per unit volume. Equations (4.11) and (4.12) are used to find the deformation pattern for which the first derivative of the adiabatic potential vanishes:

$$\Delta_{\text{eq}}(\mathbf{u}) = \frac{-\int d\mathbf{r} |\phi_n(\mathbf{r}, \Delta_{\text{eq}})|^2 Z(\mathbf{r} - \mathbf{u})}{S}. \quad (4.13)$$

This relation provides an implicit equation for the equilibrium deformation surrounding the electron. The equilibrium deformation surrounding the carrier depends on (1) the continuum's stiffness, (2) the magnitude and range of the electron–phonon interaction, and (3) the self-trapped electron's spatial extent.

The electron also alters the stiffness associated with deforming the medium surrounding it. Elements of the stiffness tensor linking deformations $\Delta(\mathbf{u})$ and $\Delta(\mathbf{u}')$ in the presence of the added electron, $S_e(\mathbf{u}, \mathbf{u}')$, are given by second partial derivatives of the adiabatic potential with respect to these deformation parameters:

$$S_e(\mathbf{u}, \mathbf{u}') = S\delta(\mathbf{u} - \mathbf{u}') + \int d\mathbf{r} \frac{\partial |\phi_n(\mathbf{r}, \Delta)|^2}{\partial \Delta(\mathbf{u}')} Z(\mathbf{r} - \mathbf{u}). \quad (4.14)$$

Equation (4.14) shows that carrier-induced changes of the system's stiffness result from the wavefunction of an electron changing as the continuum is deformed. The wavefunction changes because deforming the continuum alters the potential well experienced by the electron. Figure 4.1 illustrates the adjustment of a localized electron in response to displacing a solid's ions from their minimum-energy configuration. The electron adjusts as the ions shift so as to minimize its energy. Since this electronic relaxation lowers the net energy required to produce the deformation, the associated stiffness is reduced. This phenomenon is termed *carrier-induced softening* (Emin, 1994a; Emin, 1994b; Emin, 2000a).

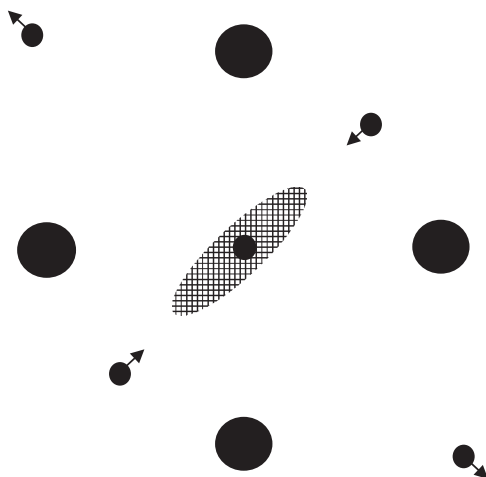


Fig. 4.1 Asymmetrical displacements of cations from their equilibrium positions (small dots with arrows) asymmetrically deform the charge distribution of the self-trapped electron (hatched region) that occupies the central cation. The large dots are anions that remain fixed at their equilibrium positions.

This carrier-induced softening may be explicitly expressed in terms of the electron–phonon interaction by using second-order perturbation theory to describe the deformation-induced change of the electronic wavefunction:

$$S_e(\mathbf{u}, \mathbf{u}') = S\delta(\mathbf{u} - \mathbf{u}') - \sum_{n' \neq n} \frac{\int d\mathbf{r}' \phi_n^*(\mathbf{r}') Z(\mathbf{r}' - \mathbf{u}') \psi_{n'}(\mathbf{r}') \int d\mathbf{r} \psi_{n'}^*(\mathbf{r}) Z(\mathbf{r} - \mathbf{u}) \phi_n(\mathbf{r}) + \text{c.c.}}{E_{n'} - W_n}. \quad (4.15)$$

Here W_n and ϕ_n designate the energy and electronic wavefunction for the ground state while $E_{n'}$ and $\psi_{n'}$ refer to the energies and corresponding wavefunctions of excited electronic states. Carrier-induced softening is evident since $E_{n'} > W_n$. Equation (4.15) explicitly manifests the symmetry of the stiffness tensor with respect to interchange of $\mathbf{u}$ and $\mathbf{u}'$.

Carrier-induced stiffness changes can also be related to the self-trapped electron's polarizability tensor. Inserting expansions of the electron–phonon interaction function about the electron's centroid, $\mathbf{r}_c$:

$$\begin{aligned} Z(\mathbf{r} - \mathbf{u}) &\cong Z(\mathbf{r}_c - \mathbf{u}) + \frac{\partial Z(\mathbf{r}_c - \mathbf{u})}{\partial \mathbf{r}_c} \cdot (\mathbf{r} - \mathbf{r}_c) \\ &= Z(\mathbf{r}_c - \mathbf{u}) - \frac{\partial Z(\mathbf{r}_c - \mathbf{u})}{\partial \mathbf{u}} \cdot (\mathbf{r} - \mathbf{r}_c) \end{aligned} \quad (4.16)$$

into Eq. (4.15) enables it to be rewritten as

$$S_e(\mathbf{u}, \mathbf{u}') = S\delta(\mathbf{u} - \mathbf{u}') - \frac{\partial Z(\mathbf{r}_c - \mathbf{u}')}{\partial \mathbf{u}'} \cdot \overset{\leftrightarrow}{P} \cdot \frac{\partial Z(\mathbf{r}_c - \mathbf{u})}{\partial \mathbf{u}}, \quad (4.17)$$

where the electronic polarizability tensor is

$$\overset{\leftrightarrow}{P} \equiv \sum_{n' \neq n} \frac{\int d\mathbf{r}' \phi_n^*(\mathbf{r}') (\mathbf{r}' - \mathbf{r}_c) \psi_{n'}(\mathbf{r}') \int d\mathbf{r} \phi_{n'}^*(\mathbf{r}) (\mathbf{r} - \mathbf{r}_c) \psi_n(\mathbf{r}) + \text{c.c.}}{E_{n'} - W_n}. \quad (4.18)$$

The electronic polarizability tends to increase strongly with the size of the electronic state. A simple model gives the polarizability increasing in proportion to the fourth power of the state's radius. As a result, carrier-induced softening may be especially important in molecular solids when an electronic state is comparable in size to a molecule.

Nonetheless, a carrier confined to a single molecule of Holstein's pedagogic molecular-crystal model does not soften its vibrations (Holstein, 1959a). Rather the model implicitly represents the localized carrier as a non-polarizable fixed point. Indeed, the model's pseudo-molecule is designed to depict the situation envisioned in simple transition-metal oxides: large displaceable anions surrounding a small fixed cation that can be occupied by an electronic hole.

The ground-state values of the electronic energy and wavefunction, W_n and $\phi_n(\mathbf{r})$, are obtained when the continuum deforms to its carrier-induced equilibrium configuration. In particular, the potential in the Hamiltonian for the ground-state eigenvalue equation is that produced by the equilibrium deformation of Eq. (4.13). Since the equilibrium deformation depends on the electron's wavefunction, the resulting eigenvalue equation is nonlinear:

$$\left[\frac{-\hbar^2}{2m} \nabla_r^2 - \frac{1}{S} \int d\mathbf{r}' |\phi_n(\mathbf{r}')|^2 \int d\mathbf{u} Z(\mathbf{r} - \mathbf{u}) Z(\mathbf{r}' - \mathbf{u}) \right] \phi_n(\mathbf{r}) = W_n \phi_n(\mathbf{r}). \quad (4.19)$$

Solutions of this differential equation depend qualitatively on the range of the electron-phonon interaction and on the electron's dimensionality. That is, the solution depends on whether the electron can move in all three dimensions or is restricted to a plane or to a chain. These features are discussed in Section 4.3.

The energy of a polaron can be written as the sum of three contributions: (1) the ground-state's electronic energy, (2) the associated deformational energy, and (3) the change of vibrations' free energy resulting from carrier-induced alterations of their frequencies. The ground-state's electronic energy is found from Eq. (4.19) to be:

$$W_n = \frac{\hbar^2}{2m} \int d\mathbf{r} |\nabla_r \phi_n(\mathbf{r})|^2 - \frac{1}{S} \int d\mathbf{r} |\phi_n(\mathbf{r})|^2 \int d\mathbf{r}' |\phi_n(\mathbf{r}')|^2 \int d\mathbf{u} Z(\mathbf{r} - \mathbf{u}) Z(\mathbf{r}' - \mathbf{u}). \quad (4.20)$$

The strain energy needed to produce the ground-states' equilibrium deformation is

$$\frac{S}{2} \int d\mathbf{u} \Delta_{\text{eq}}^2(\mathbf{u}) = \frac{1}{2S} \int d\mathbf{r} |\phi_n(\mathbf{r})|^2 \int d\mathbf{r}' |\phi_n(\mathbf{r}')|^2 \int d\mathbf{u} Z(\mathbf{r} - \mathbf{u}) Z(\mathbf{r}' - \mathbf{u}). \quad (4.21)$$

This deformation energy has the same form as the deformation-induced lowering of the electron's potential energy. Thus these two contributions are combined in writing the minimum of the adiabatic potential of Eq. (4.5) with $\Delta \equiv \Delta_{\text{eq}}$ as:

$$V_{\text{ad}}^{\text{min}} = \frac{\hbar^2}{2m} \int d\mathbf{r} |\nabla_r \phi_n(\mathbf{r})|^2 - \int d\mathbf{r} |\phi_n(\mathbf{r})|^2 \int d\mathbf{r}' |\phi_n(\mathbf{r}')|^2 I(\mathbf{r}, \mathbf{r}'), \quad (4.22)$$

where

$$I(\mathbf{r}, \mathbf{r}') \equiv \frac{1}{2S} \int d\mathbf{u} Z(\mathbf{r} - \mathbf{u}) Z(\mathbf{r}' - \mathbf{u}). \quad (4.23)$$

The interaction function $I(\mathbf{r}, \mathbf{r}')$ has been evaluated for models of the electron–phonon interaction. For the Fröhlich long-range electron–phonon interaction the interaction function is

$$I_{\text{LR}}(\mathbf{r}, \mathbf{r}') = \left(\frac{1}{\varepsilon_\infty} - \frac{1}{\varepsilon_0} \right) \frac{e^2}{2|\mathbf{r} - \mathbf{r}'|}. \quad (4.24)$$

The interaction function for the short-range electron–phonon interaction of Eq. (2.4) is readily calculated to yield

$$I_{\text{SR}}(\mathbf{r}, \mathbf{r}') = \frac{F^2}{2k} V_c \delta(\mathbf{r} - \mathbf{r}') \quad (4.25)$$

where $S \equiv k/V_c$ and V_c is the unit cell volume. Upon summing these long-range and short-range components of the interaction function, the adiabatic potential, Eq. (4.22), becomes

$$V_{\text{ad}}^{\text{min}} = \frac{\hbar^2}{2m} \int d\mathbf{r} |\nabla_{\mathbf{r}} \phi_n(\mathbf{r})|^2 - \frac{e^2}{2} \left(\frac{1}{\varepsilon_\infty} - \frac{1}{\varepsilon_0} \right) \int d\mathbf{r} \int d\mathbf{r}' \frac{|\phi_n(\mathbf{r})|^2 |\phi_n(\mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|} - \left(\frac{F^2}{2k} \right) V_c \int d\mathbf{r} |\phi_n(\mathbf{r})|^4. \quad (4.26)$$

The third contribution to a polaron's energy is produced by its altering of the frequencies with which the atoms surrounding it oscillate. The associated change of the free energy of these atoms' vibrations is

$$\Delta F_{\text{vib}} \equiv \kappa T \sum_i \ln \frac{\sinh[\hbar(\omega_i - \Delta\omega_i)/2\kappa T]}{\sinh(\hbar\omega_i/2\kappa T)}, \quad (4.27)$$

where deformation-induced polarization of the electronic carrier shifts the i -th vibration frequency from ω_i to $\omega_i - \Delta\omega_i$. Here κ is the Boltzmann constant and T is the absolute temperature. At absolute zero the carrier-induced change of vibrations' free energy is just the net carrier-induced shift of vibrations' zero-point energy:

$$\Delta F_{\text{vib}} = -\frac{\hbar}{2} \sum_i \Delta \omega_i. \quad (4.28)$$

At finite temperatures the reduction of vibrations' free energy can be approximated by

$$\Delta F_{\text{vib}} \cong -\sum_i \frac{\hbar \Delta \omega_i}{2} \coth(\hbar \omega_i / 2\kappa T), \quad (4.29)$$

when the carrier-induced changes of vibration frequencies are sufficiently small.

4.3 Scaling approach: large- and small-polaron formation

Adiabatic studies of self-trapping usually begin by passing to the limit of large atomic masses. Vibration frequencies then vanish as atoms become infinitely massive with the medium's stiffness remaining finite. Phenomena associated with atomic movements are thereby ignored. In particular, (1) carrier-induced softening of atoms' vibrations, (2) polaron motion and (3) non-adiabatic transitions are suppressed. In this limit a polaron's energy becomes just the adiabatic potential of Eq. (4.22). A self-trapped carrier's wavefunction is that which minimizes this energy.

A scaling analysis of the minimized adiabatic potential ground state yields its fundamental features without requiring explicit solution of the nonlinear differential equation, Eq. (4.19). The scaling analysis considers the effect on the energy of the adiabatic ground state of changing the length-scale of its electronic wavefunction (Emin and Holstein, 1976). In particular, $\mathbf{r}$ is replaced by $\mathbf{r}/L$ in $\phi_n(\mathbf{r})$ while its normalization is maintained by multiplying it by $L^{-d/2}$, where d is the electron's dimensionality. The energy functional obtained by this replacement, $E(L)$, is then studied as a function of the dimensionless scaling factor, L . Altering the length-scale of the wavefunction of the adiabatic ground state necessarily increases the corresponding energy. Therefore, the minimum of the energy functional occurs at $L = 1$.

Applying the scaling procedure to the adiabatic potential of Eq. (4.22) (plus changing integration variables) ultimately yields the energy functional:

$$E(L) = \frac{T_e}{L^2} - \frac{V_L}{L} - \frac{V_S}{L^d}, \quad (4.30)$$

where

$$T_e \equiv \frac{\hbar^2}{2m} \int d\mathbf{r} |\nabla_r \phi_n(\mathbf{r})|^2, \quad (4.31)$$

$$V_L \equiv \frac{e^2}{2} \left(\frac{1}{\varepsilon_\infty} - \frac{1}{\varepsilon_0} \right) \int d\mathbf{r} \int d\mathbf{r}' \frac{|\phi_n(\mathbf{r})|^2 |\phi_n(\mathbf{r}')|^2}{|\mathbf{r} - \mathbf{r}'|}, \quad (4.32)$$

and

$$V_S \equiv \left(\frac{F^2}{2k} \right) V_c \int d\mathbf{r} |\phi_n(\mathbf{r})|^4. \quad (4.33)$$

As shown in Fig. 4.2, with only the long-range electron–phonon interaction the energy functional $E(L)$ has a single finite-energy minimum. This minimum occurs at $L = 2T_e/V_L$ where $E(L) = -V_L^2/4T_e$. Net binding occurs because the lowering of the carrier’s energy, $T_e/L^2 - 2V_L/L = -3V_L^2/4T_e$, is 3/2 of the associated deformation energy, $V_L/L = V_L^2/2T_e$. The requirement that $L = 1$ at the minimum insures that $2T_e = V_L$ for the ground state.

A qualitatively different situation prevails with only the short-range electron–phonon interaction. As depicted in Fig. 4.3, the energy functional with full ($d = 3$) electronic dimensionality has minima at $L = 0$ and at $L = \infty$ separated by a maximum at $L = 3V_S/2T_e$ (Emin, 1973a). These two minima respectively correspond to a self-trapped carrier that collapses to a severely localized state and to a free carrier that extends over all space. The maximum, $8T_e^3/27V_S^2$, is termed the “barrier to self-trapping” because a free carrier must transcend this barrier in order to self-trap (Rashba, 1957; Toyozawa, 1961). As such, a free carrier experiences a time delay before self-trapping (Mott and Stoneham, 1977). In particular, *persistent photo-conductivity*

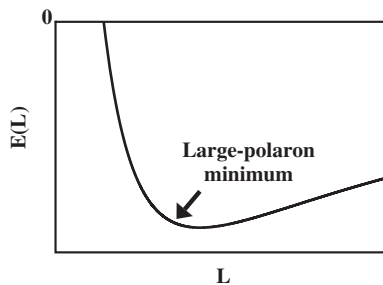


Fig. 4.2 The finite-radius of the minimum of the adiabatic energy functional $E(L)$ for a carrier in a three-dimensional ionic medium indicates that its self-trapped carrier forms a large polaron.

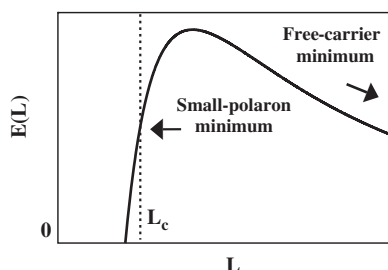


Fig. 4.3 No well-defined minimum exists for the adiabatic energy functional $E(L)$ of a carrier interacting with a three-dimensional continuum via a short-range electron–phonon interaction. A carrier either remains free, $E(L) \rightarrow 0$ as $L \rightarrow \infty$, or the carrier self-traps as a small polaron: $E(L) \rightarrow -\infty$ as $L \rightarrow 0$. The shrinking of a carrier is limited to L_c by a real material’s atomicity. Small-polaron formation is indicated by $\partial E(L)/\partial L > 0$ at L_c .

occurs when a photo-carrier moves freely for a long time before it collapses into an energetically favorable severely localized state whose energy is significantly modified by the short-range electron–phonon interaction (Henry and Lang, 1977).

The three-dimensional continuum model with the short-range electron–phonon interaction gives $E(L) \rightarrow -\infty$ as $L \rightarrow 0$. This behavior describes a carrier becoming self-trapped at an energy that becomes infinitely deep as the carrier’s potential well becomes infinitesimally small. However, beyond the continuum model in a real material this self-trapping *collapse* actually saturates as the size of the potential well shrinks to atomic dimensions. This saturation can be described by imposing a small- L cut-off on $E(L)$ at the value $L = L_c$. The value of $E(L_c) \equiv E_{sp}$ is the energy of a *small* polaron. By contrast, an unsaturated finite-radius polaron minimum, as found with the three-dimensional long-range electron–phonon interaction, indicates formation of a *large* polaron.

In the adiabatic limit, the small-polaron minimum at $E(L_c)$ coexists with the free-carrier minimum at $E(\infty)$. The small-polaron minimum is energetically stable if $E(L_c) < E(\infty) = 0$. In terms of microscopic parameters, a small-polaron’s energetic stability requires that the lowering of its potential energy, $F^2/2k$, exceed the energy required to confine an electronic carrier to a site, essentially half the electronic bandwidth, $W/2 = 6t$ for a tight-binding band in a cubic crystal.

The dichotomy between a carrier remaining free and collapsing to form a small-polaron persists if the carrier’s motion is limited to a plane ($d=2$) with a short-range interaction. In particular, the solution is severely localized if $V_S > T_c$ and is delocalized if $V_S < T_c$. A type of two-dimensional self-trapping, electronically stimulated desorption from a surface, will be addressed in Chapter 16.

Self-trapping in quasi-one-dimensional materials has received considerable attention. Analysis of Eq. (4.30) reveals that a finite-radius solution occurs for a

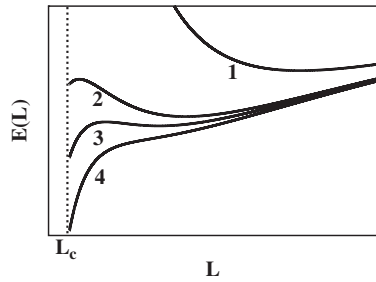


Fig. 4.4 The large-polaron minimum (curve 1) shrinks and falls in energy as the short-range interaction increases. As the short-range interaction strengthens (curve 2) a meta-stable small-polaron minimum appears at L_c . The small polaron is stabilized as the interaction is increased further (curve 3). Ultimately (curve 4) the large-polaron minimum disappears.

carrier whose motion is limited to an isolated chain ($d = 1$). The energy function then has its solitary minimum at $L = 2T_e/V_S$ where $E(L) = -V_S^2/4T_e$. Explicit solution for a one-dimensional chain yields a ground-state energy of $-t/3L_p^2$ for a self-trapped state whose half-length encompasses $L_p = 2t/(F^2/2k)$ chain units with an inter-unit electronic transfer energy of t (Holstein, 1959a). However, self-trapping in a quasi-one-dimensional anisotropic material requires severe localization in the two directions perpendicular to the chain. In particular, such localization costs about $2(2t_t)$, the half-bandwidth $2t_t$ in each of the two directions transverse to the chain. Thus the binding energy of a quasi-one-dimensional polaron is $-t/3L_p^2 + 4t_t$. Stability of a quasi-one-dimensional large polaron with respect to its carrier remaining free therefore requires very strong anisotropy of the electronic transfer energies: $t/t_t > 12L_p^2$ (Emin, 1986; Schüttler and Holstein, 1986).

Often a carrier interacts both with displacements of distant ions and with those of the atoms it contacts. The energy functional Eq. (4.30) describes both long- and short-range interactions. For a carrier confined to a chain ($d = 1$) embedded within an ionic medium the contributions to $E(L)$ from both interactions are simply summed as both are proportional to $1/L$. For a carrier confined to a plane ($d = 2$), the short-range interaction is proportional to $1/L^2$. As such, its effect is to simply replace the coefficient of the confinement-energy contribution T_e by $T_e - V_S$. With $d = 3$ long- and short-range electron-phonon interactions combine to produce the four possible polaron scenarios depicted in Fig. 4.4. As illustrated in curve 1, only a large polaron can form for a weak short-range interaction. Increasing the strength of the short-range interactions facilitates formation of a small polaron that may either be meta-stable or stable with respect to the large-polaron minimum, curves 2 and 3, respectively. As shown in curve 4, only small-polaron formation is possible with a sufficiently strong short-range interaction: $V_S > T_e^2/3V_L$. In particular, the long-range electron-phonon interaction facilitates a carrier's collapse into a small polaron.

4.4 Self-trapping beyond the adiabatic limit

Beyond the adiabatic limit (infinitely massive atoms) atoms can vibrate. Atomic vibrations enable both carrier-induced softening and strong-coupling polarons' motion. Atoms' vibrations affect polarons' energies, stability and their ability to form. In particular, the small-polaron and free-carrier coexistence domain contracts as atoms' vibration frequencies increase.

The potential experienced by a self-trapped carrier changes as atoms vibrate. The adjustment of a self-trapped carrier to this changing potential alters the stiffness constants and vibration frequencies associated with these atomic vibrations. If the carrier-induced shifts of atomic frequencies are sufficiently small the change of the free energy of atoms' vibrations is given by Eq. (4.27). For dispersion-less optic vibrations of frequency ω the change of vibrations' free-energy just depends on the net shift of atoms' vibration frequencies:

$$\Delta F_{\text{vib}} \cong -\frac{\hbar \coth(\hbar\omega/2\kappa T)}{2} \sum_i \Delta\omega_i. \quad (4.34)$$

Furthermore, the sum of the carrier-induced shifts of the optic frequencies is simply related to the sum of diagonal ($\mathbf{u} = \mathbf{u}'$) carrier-induced stiffness changes (Emin, 2000a):

$$\begin{aligned} \sum_i \Delta\omega_i &= \frac{\omega}{S} \sum_{n' \neq n} \frac{\int d\mathbf{r}' \phi_n^*(\mathbf{r}') \psi_{n'}(\mathbf{r}') \int d\mathbf{r} \psi_{n'}^*(\mathbf{r}) \phi_n(\mathbf{r}) \int d\mathbf{u} Z(\mathbf{r}' - \mathbf{u}) Z(\mathbf{r} - \mathbf{u})}{E_{n'} - W_n} \\ &= 2\omega \sum_{n' \neq n} \frac{\int d\mathbf{r}' \rho_{n,n'}^*(\mathbf{r}') \int d\mathbf{r} \rho_{n,n'}(\mathbf{r}) I(\mathbf{r}, \mathbf{r}')}{E_{n'} - W_n}, \end{aligned} \quad (4.35)$$

where Eqs. (4.15) and (4.23) have been utilized and the overlap density has been defined by

$$\rho_{n,n'}(\mathbf{r}) \equiv \psi_{n'}^*(\mathbf{r}) \phi_n(\mathbf{r}). \quad (4.36)$$

Combining Eqs. (4.34) and (4.35) yields the carrier-induced change of the free energy of dispersion-less optic vibration for a self-trapped carrier in state n :

$$\Delta F_{\text{vib}} \cong -\hbar\omega \coth(\hbar\omega/2\kappa T) \sum_{n' \neq n} \frac{\int d\mathbf{r}' \rho_{n,n'}^*(\mathbf{r}') \int d\mathbf{r} \rho_{n,n'}(\mathbf{r}) I(\mathbf{r}, \mathbf{r}')}{E_{n'} - W_n}. \quad (4.37)$$

This carrier-induced shift of atomic-vibrations' free-energy contributes to the polaron energy and thereby affects its stabilization. In the zero-temperature limit this shift is just that from atoms' zero-point vibrations, proportional to $\hbar\omega$. The shift tends to be largest for very large self-trapped states as occur for molecular polarons (Emin, 2000a).

In the adiabatic limit, where atomic vibrations are suppressed, self-trapped carriers are strictly localized. Beyond the adiabatic limit, however, atoms vibrate and self-trapped carriers can move. Concomitantly, a strong-coupling polaron's mass falls from infinity in the adiabatic limit to a finite value that generally greatly exceeds that of a free electronic carrier. The associated polaron bandwidth rises from zero in the adiabatic limit ($\omega = 0$) to a value that is less than the characteristic phonon energy.

A large polaron's effective mass is a weighted measure of the masses of the ions whose shifts transfer the potential well containing the self-trapped carrier (Schüttler and Holstein, 1986; Emin and Hillery, 1989). Elements of the effective mass tensor can be written as the sum of the product of the masses of displaced ions' and their self-trapping strain fields in each of two principal directions (Emin and Hillery, 1989, p. 2388). In an isotropic medium a large polaron's scalar effective mass is often simply expressed in terms of the bare carrier mass m and the Fröhlich coupling constant, α of Eq. (3.3) by (Fröhlich, 1963):

$$m_{\text{LP}} \approx m\alpha^4. \quad (4.38)$$

The strong-coupling condition $\alpha \gg 1$ ensures that the large-polaron effective mass greatly exceeds the bare carrier mass. Using the definition of α along with expressions for the binding energy $E_{\text{LP}} = V_{\text{L}}^2/4T_{\text{e}}$ and spatial extent $R_{\text{LP}} = (2T_{\text{e}}/V_{\text{L}})a$ of the large polaron, its effective mass can be rewritten as (Emin and Hillery, 1989; Emin, 1993):

$$m_{\text{LP}} \approx \frac{E_{\text{LP}}}{\omega^2 R_{\text{LP}}^2}, \quad (4.39)$$

where $m_{\text{LP}} \rightarrow \infty$ as $\omega \rightarrow 0$, the adiabatic limit. Knowing this mass enables estimation of the kinetic energy at which a large-polaron's de Broglie wavelength falls to its characteristic spatial extent $\sim R_{\text{LP}}$:

$$E_{\text{c}} \approx \frac{(\hbar/R_{\text{LP}})^2}{m_{\text{LP}}} \approx \hbar\omega \left(\frac{\hbar\omega}{E_{\text{LP}}} \right). \quad (4.40)$$

Noting the self-trapping condition, $E_{\text{LP}} > \hbar\omega$, it becomes evident that $E_{\text{c}} < \hbar\omega$.

Small-polaron motion is qualitatively different from large-polaron motion. As illustrated in the two-site example of Section 3.3, an energy barrier is encountered as a small polaron moves between sites. Since the change of a carrier's electronic energy as it passes over a barrier exceeds its electronic transfer energy, a carrier loses coherence in such transfer processes. Coherent small-polaron transport is nonetheless possible with direct tunneling through this energy barrier. Such tunneling exchanges the pattern of atomic deformations about a site occupied by a small polaron with the atomic-deformation pattern surrounding an unoccupied site. These atomic tunneling processes are also described as “diagonal transitions” in which atomic displacements are exchanged between sites without phonons being emitted or absorbed (Holstein, 1959b, pp. 348–355). The width of a small-polaron band is a measure of the energy splitting produced by this tunneling. As such, small-polaron bandwidths vanish in both classical and adiabatic limits, $\hbar \rightarrow 0$ and $\omega \rightarrow 0$, since such tunneling is then suppressed.

Consider tunneling between the atomic deformation patterns of the two minima of the ground-state adiabatic potential Eq. (3.14) plotted in Fig. 4.5. The energy splitting of the adiabatic ground state is twice the matrix element of the Hamiltonian between local functions centered at different minima. Ignoring the t -dependent rounding of the barrier and approximating the mutually orthogonal local functions as displaced-harmonic-oscillator ground-state wavefunctions, the tunnel splitting is seen to be much less than the phonon energy $\hbar\omega$:

$$2\langle X(x \pm F/\sqrt{2k}) | H | X(x \mp F/\sqrt{2k}) \rangle \cong \hbar\omega \left[\sqrt{\frac{(F^2/2k)}{\hbar\omega}} e^{-(F^2/2k)/\hbar\omega} \right], \quad (4.41)$$

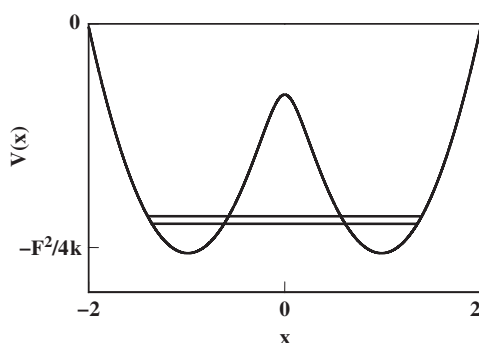


Fig. 4.5 With small-polaron formation the lowest energy levels of the two-site ground-state adiabatic potential reside at an energy of $\cong \hbar\omega/2$ above the minima at $\approx -F^2/4k$. Tunneling between these two minima splits these levels' degeneracy thereby forming an energy doublet which is represented by the figure's two horizontal lines.

where $F^2/2k \gg \hbar\omega$. The ground-state tunnel splitting becomes even smaller, $\approx 2t \exp[-(F^2/2k)/(\hbar\omega)]$, if the electronic transfer energy is small enough to invalidate the adiabatic approach, $2t \ll [(F^2/2k)(\hbar\omega)]^{1/2}$ (Holstein, 1959b, pp. 370–375).

Overlap between displaced harmonic oscillators produces the exponential reduction factor of the small-polaron tunnel splitting. Increasing the energies of these harmonic oscillators' vibrations increases the spatial extents of their wavefunctions. However, the wiggles of the wavefunction of a harmonic oscillator increase with its energy. As a result the net overlap between displaced harmonic oscillators decreases as their energies are increased. Thus the thermally averaged overlap between displaced oscillators falls with increasing temperature. In particular, the exponential overlap factor is reduced to $\exp\{-(F^2/2k)/\hbar\omega\} \coth(\hbar\omega/2kT)$ at finite temperatures.

Small-polaron bands are typically too narrow to support coherent motion. For example, the adiabatic tunnel splitting of Eq. (4.41) is < 0.01 eV for even very modest values of the parameters: $\hbar\omega = 0.1$ eV and $(F^2/2k)/\hbar\omega = 3$. For $\hbar\omega = 0.03$ eV and $(F^2/2k)/\hbar\omega = 10$ the adiabatic tunnel splitting is $< 10^{-5}$ eV. Thus, the site-to-site disorder energy produced by charge centers, defects and even applied fields often exceeds the small-polaron bandwidth. Coherent motion is then destroyed and small polarons move incoherently via thermally assisted hopping as described in Chapter 11.

Variational studies indicate that a carrier added to a three-dimensional deformable crystal either remains nearly free or collapses to form a small polaron (Toyozawa, 1961; Emin, 1973a). As depicted in the uppermost panel of Fig. 4.6, consistent with the results of Section 4.3 for the adiabatic limit ($\omega = 0$), the zero-momentum ground states of these two solutions coexist. The free carrier is energetically stable when $F^2/2k < W/2$ and the small-polaron solution is energetically stable when $F^2/2k > W/2$. As shown in the middle panel of Fig. 4.6, beyond the adiabatic limit, as $\hbar\omega/(W/2)$ rises from zero, the values of $(F^2/2k)/(W/2)$ between which these two solutions coexist shrinks. In particular, a small polaron cannot form for too weak an electron–phonon coupling and a carrier cannot exist as nearly free for too strong an electron–phonon coupling. The energy of the small-polaron's zero-momentum state is given by

$$E_{\text{SP}} = -z\hbar\omega \sqrt{\left(\frac{F^2/2k}{\hbar\omega}\right)} \exp\left(-\frac{F^2/2k}{\hbar\omega}\right) - F^2/2k, \quad (4.42)$$

where z is the number of nearest neighbors ($z = 6$ for a cubic lattice) and small-polaron transfer is taken as adiabatic. The free-carrier energy is approximately that obtained by treating the electron–phonon interaction as a small perturbation:

$$E_{\text{FC}} = -W/2 - \frac{(F^2/2k)\hbar\omega}{W/2}. \quad (4.43)$$

As depicted in the lowest panel of Fig. 4.6, the coexistence regime disappears when the electronic bandwidth divided by the phonon energy is small enough. Then these variational treatments fail between their free-carrier and small-polaron limits.

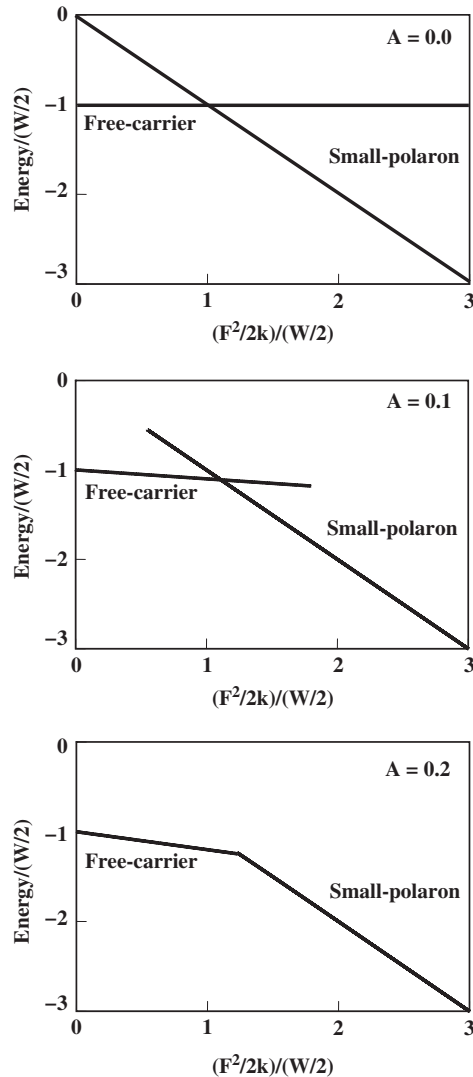


Fig. 4.6 The free-carrier and small-polaron ground-state energies in units of half the electronic bandwidth $W/2$ are plotted against the electron–phonon coupling strength $(F^2/2k)/(W/2)$ for three values of $A \equiv \hbar\omega/(W/2)$. The coupling strengths between which both solutions coexist diminishes as A increases from 0, the adiabatic limit.

To understand the condition limiting small-polaron formation, consider the energy functional for the energy of a zero-momentum small-polaron divided by $F^2/2k$:

$$\frac{E(y)}{F^2/2k} = -z \frac{\exp\left(-\frac{F^2/2k}{\hbar\omega} y^2\right)}{\sqrt{\left(\frac{F^2/2k}{\hbar\omega}\right)}} - 2y + y^2, \quad (4.44)$$

where y is a dimensionless measure of the displacements of the equilibrium positions of atoms surrounding the carrier. The first contribution on the right-hand side of Eq. (4.44) is proportional to the lowering of the small-polaron's energy due to its adiabatic tunneling to adjacent sites. The second and third contributions on the right-hand side of Eq. (4.44) are proportional to the lowering of the carrier's energy and the associated increase of the solid's strain energy, respectively. With small-polaron formation this functional has a minimum near $y = 1$. However, as shown in Fig. 4.7, this small-polaron minimum disappears when the coupling strength $(F^2/2k)/\hbar\omega$ becomes small enough for the tunneling contribution to the small-polaron energy to become significant.

The electron–phonon coupling strength becomes too strong for a carrier to move freely with its interaction with the phonons being just a perturbation when the two contributions to E_{FC} of Eq. (4.43) become comparable to one another. This condition can be expressed as:

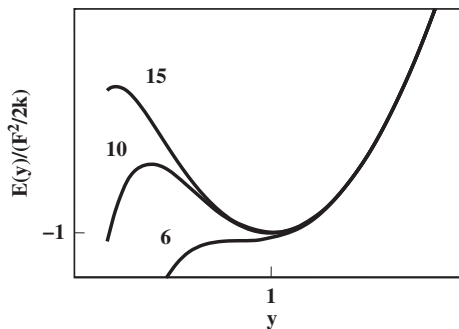


Fig. 4.7 The energy functional $E(y)$ in units of $F^2/2k$ is plotted against the deformation parameter y for four values of the electron–phonon coupling parameter $(F^2/2k)/\hbar\omega$. The minimum of $E(y)$ at $y \approx 1$, indicating small-polaron formation, vanishes when $(F^2/2k)/\hbar\omega$ falls below 6.

$$F\left(\frac{\hbar}{W/2}\right) > \sqrt{M\hbar\omega}. \quad (4.45)$$

Thus, the carrier can no longer move freely if the product of the force it exerts on the surrounding atoms F multiplied by the time that a free carrier can be held at a site $2\hbar/W$ exceeds atoms' zero-point momentum $\sqrt{M\hbar\omega}$ (Emin, 1973a). That is, a carrier can no longer move freely if it lingers at a site long enough for its presence to significantly alter atoms' motion.

So far this discussion has only considered a carrier interacting with zero-point phonons. However, the variational calculations and the physical arguments governing the validity of the small-polaron and free-carrier solutions survive extension to finite temperature (Emin, 1972; Emin 1973a). As the temperature is raised the domain in which small-polaron and free-carrier solutions coexist widens. This situation suggests a type of thermally driven conductivity transition. In particular, below the transition temperature all carriers self-trap and have low mobility while above the transition temperature conduction is dominated by high-mobility free carriers (Emin, 1972; Emin 1973a).

4.5 Disorder-assisted small-polaron formation

Evidence of small polarons is found among non-crystalline solids. In these instances the absence of crystallinity presents charge carriers with a disordered environment. By itself, disorder fosters electronic localization and makes extended-state motion tortuous (Anderson, 1958). Since confining and slowing a charge carrier facilitates the displacing of surrounding atoms, disorder aids a carrier's collapse into a small polaron. By driving a moderately localized or tenuously moving carrier to collapse into a severely localized self-trapped state, electron-phonon interactions amplify the localizing tendencies of disorder. For example, time-of-flight measurements indicate that the high-mobility quasi-free motion of holes in crystalline orthorhombic sulfur collapses to low-mobility small-polaron hopping motion as this molecular crystal is heated into its "softening regime" where the S_8 molecules relinquish the regularity of their locations with respect to one another (Ghosh and Spear, 1968).

The electron-phonon interaction deepens the potential well of a localized carrier. The deepening of the carrier's potential well in turn increases the severity of the carrier's localization. This regenerative nature of self-trapping is embodied in the nonlinearity of the differential equation governing self-trapping, Eq. (4.19). As a result, sufficiently localized states in covalent glasses are expected to collapse into small polarons (Anderson, 1972).

The scaling treatment of self-trapping in the adiabatic limit of Section 4.3 can be employed to show how increasing confinement causes a localized carrier to collapse into a small polaron (Emin and Holstein, 1976). The localized state's energy functional is analogous to Eq. (4.30):

$$E(L) = \frac{T_e}{L^2} - \frac{V_S}{L^3} - \frac{V_D}{L^s}, \quad (4.46)$$

where L is the length-scaling factor and the energetic constants T_e and V_S are defined by Eqs. (4.31) and (4.33). The disorder constant associated with the localized state's bare electronic potential, $-V_0/r^s$, is given by:

$$V_D \equiv \int d\mathbf{r} |\phi_n(\mathbf{r})|^2 \left(\frac{V_0}{r^s} \right). \quad (4.47)$$

The energy functional of Eq. (4.46) is plotted in Fig. 4.8 for $s = 1$. The parameters are chosen such that the carrier is delocalized in the absence of the localizing potential, $V_D = 0$. Increasing the strength of the localizing potential introduces a meta-stable small-polaron state. Further strengthening of the localizing potential stabilizes the small-polaron state. A still stronger localizing potential eliminates all but the small-polaron solution. Thus, sufficiently localized carriers are converted into small polarons.

In a crystal the electronic eigenstates are extended with uniform charge density. Imposing disorder introduces localized states while rendering the charge densities of extended states non-uniform (Mott and Davis, 1979). The energy separating an energy band's localized states from its extended states is termed its mobility edge. The Emin–Holstein scaling analysis has been used to argue that the electron–phonon

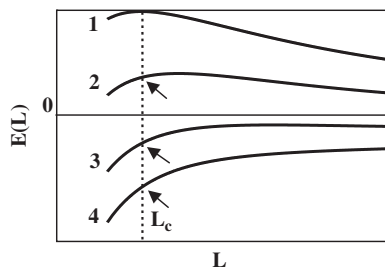


Fig. 4.8 Imposition of a localizing potential can induce a free carrier in a uniform covalent medium (curve 1) to form a small-polaron (arrows). As a localizing potential is strengthened a meta-stable small-polaron is introduced (curve 2) and then stabilized (curve 3). A strong enough potential triggers a carrier's collapse into a localized small polaron (curve 4).

interaction causes sufficiently non-uniform extended states above the mobility edge to behave analogously to localized states by collapsing into small polarons (Cohen, Economou and Soukoulis, 1983).

Another approach to disorder-induced collapse of extended-state carriers into small polarons extends the adiabatic theory of small-polaron formation in a crystal to a covalent glass (Emin and Bussac, 1994). A stable carrier in a crystal will remain free or form a small polaron depending on the magnitude of the electronic bandwidth W . Within the tight-binding model the electronic half-bandwidth is the product of the number of nearest neighbors z and the magnitude of the electronic transfer energy t : $W/2 = zt$. If $zt > F^2/2k$ a carrier moves freely between sites without polaron deformation. Alternatively, for $zt < F^2/2k$ the carrier forms a small polaron. A covalent glass is characterized by variations of its bond angles. These differences introduce significant dispersion of its electronic transfer energies. This local dispersion of electronic transfer energies inhibits a carrier's ability to avoid forming a small polaron. In particular, a carrier can only avoid forming a small polaron at site $\mathbf{g}$ if its net electronic transfer energy from that site to neighboring sites $\mathbf{g} + \mathbf{h}$ is large enough:

$$\sum_{\mathbf{h}=1}^{z'} t_{\mathbf{g}, \mathbf{g} + \mathbf{h}} > \frac{F^2}{2k}, \quad (4.48)$$

where z' indicates that the summation excludes transfers to sites at which small-polaron formation occurs. Indeed, such transfers are forbidden in the adiabatic limit $\hbar\omega \rightarrow 0$. This feature constrains a free carrier to avoid sites at which it would collapse into a small polaron. This restriction on a free carrier's movement also fosters its collapse into a small polaron. As a result of this feedback effect modest transfer-energy disorder can significantly reduce the electron–phonon coupling strength required for global small-polaron formation (Emin and Bussac, 1994). Indeed steady-state transport studies in covalent glasses indicate small-polaron hopping (Emin, Seager and Quinn, 1972; Seager, Emin and Quinn, 1973; Seager and Quinn, 1975; Klaffe and Wood, 1976; Lewis, 1976a; Lewis, 1976b; Moustakas and Paul, 1977; Baily and Emin, 2006a; Baily, Emin and Li, 2006b).

4.6 Synopsis

An electronic carrier typically adjusts to the sluggish motions of the relatively heavy atoms that surround it. As such, a carrier may find itself bound within the potential well produced by suitable atomic displacements. This atomic configuration will appear nearly static to the bound carrier if its binding energy is larger than the characteristic phonon energy. The carrier is then self-trapped, bound within a

potential well produced by surrounding atoms being displaced to new equilibrium positions consistent with the carrier's presence.

A polaron denotes the unit comprising the self-trapped carrier and the surrounding atoms at displaced equilibrium positions. The spatial extent of a large polaron's self-trapped carrier increases continuously with the magnitude of the electronic bandwidth and typically extends over multiple atomic sites. By contrast, a small polaron's self-trapped carrier collapses to the smallest relevant structural unit.

A self-trapped carrier interacting with the displaceable ions of a three-dimensional polar medium only via the long-range (Fröhlich) electron–phonon interaction will generally form a large polaron. By contrast, a carrier with only a short-range electron–phonon interaction will either remain free or collapse into a small polaron. A small polaron will be energetically stable if its binding energy $F^2/2k$ exceeds the electronic band's half-width $W/2$. Exceptionally, a short-range electron–phonon interaction supports a large polaron in a quasi-one-dimensional material if its electronic transfer energies are extremely anisotropic.

Typically a large polaron can form in an ionic solid. By contrast, by itself the short-range electron–phonon interaction is often neither strong enough nor the electronic energy bands narrow enough to produce an energetically stable small polaron $F^2/2k > W/2$. Nonetheless, adding the localizing effects of disorder and/or the long-range electron–phonon interaction frequently triggers a carrier's collapse into a small polaron.

Self-trapping is often studied within the adiabatic limit where atoms' motion is arbitrarily slow. Beyond the adiabatic limit atoms are permitted to vibrate with finite frequencies. Moreover, a self-trapped carrier's adjusting to the changing positions of atoms shifts their vibration frequencies. Atomic vibrations also enable a polaron to move and thereby garner a finite bandwidth. The width of a large-polaron band is less than the characteristic phonon energy. A small-polaron band is usually much narrower than even a large-polaron band. A large polaron moves coherently with its motion impeded by occasional scattering events. Small-polaron motion is generally incoherent via a succession of phonon-assisted hops.

Beyond the adiabatic limit a small polaron can only exist beyond a minimum value of the short-range electron–phonon coupling strength. A free carrier can only exist below a maximum value of the short-range electron–phonon coupling strength. With $W \gg \hbar\omega$ there is a span of short-range electron–phonon coupling strengths within which free carriers and small polarons can coexist. This coexistence domain increases with increasing temperature. Thus, if only small polarons form at low temperatures above a transition temperature some carriers will abruptly be freed. Below this transition temperature transport will be by low-mobility small-polaron hopping. Above the transition temperature transport may be dominated by conventional higher mobility carriers.